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ABSTRACTS AND PROGRAM

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EXPLANATION OF THE WATER MONITORING TIME SERIES DATA IN SCHOHARIE CREEK, NY

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Introduction

Flooding in the rivers in upstate New York is an event that occurs mostly either due to rainfall (tropical storms) or due to ice jams (Johnston & Garver, 2001; Lederer & Garver, 2001; Sheller et al., 2002; Garver & Cockburn, 2009; Marsellos et al., 2010a). In case of flooding, it has been observed a higher level in the water discharge of the Mohawk river and Schoharie Creek (e.g., Marsellos et al., 2010b). In this study, we analyze the time series of the daily water discharge data from the Schoharie Creek for the period of 2005-2013. In particular, we investigate the association of flooding with the climatic variables as well as the water underground level time series data derived from the same location and during the same time period.

Methodology

In several studies it has been proved that the separation of a time series into different components is necessary, to avoid contribution of different covariance structures between the components of the time series (Zurbenko & Sowizral 1999; Tsakiri & Zurbenko 2008; Tsakiri & Zurbenko 2009). For this reason, in order to examine the association of the time series of the water discharge with the climatic variables, we first decompose the time series of all the variables. A statistical methodology is presented for the decomposition of the time series into the long, seasonal and short term component of the variables. For the decomposition of the time series we use the Kolmogorov-Zurbenko filter (Zurbenko 1986). The Kolmogorov-Zurbenko (KZ) filter provides a simple design and the smallest interferences between the scales of a time series (Yang & Zurbenko 2010b). Moreover the KZ filter provides effective separation of frequencies and can be applied directly to datasets containing missing observations (Eskridge et al. 1997; Rao et al. 1997; Yang & Zurbenko, 2010a).

Daily underground water level time series data and daily climatic data have been analyzed for the same time period (Jan. 2005 to Feb. 2013) at the same location. After the application of the KZ filter, we analyze each component separately and we prove that the components of the water discharge time series can be explained by the climatic variables and the underground water level. This methodology can be applied for studying the water discharge time series in other locations, as well.

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